CLINICAL · MOUTH & THROAT

# Oral ulcers, white patches and submucous fibrosis

Most mouths that hurt have a harmless reason: an aphthous ulcer, a sharp tooth, a crop of herpes or a patch of thrush. The skill is in sorting the many benign ulcers and white patches from the few that matter, and in remembering the one rule that keeps patients alive: an ulcer that has not healed in three weeks, once the cause has been removed, is biopsied.

**AT A GLANCE**

- Aphthous ulcers recur on movable mucosa and heal in two weeks.
- A white patch that will not wipe off is not thrush: biopsy.
- Submucous fibrosis: areca nut, blanching, bands and trismus; premalignant.
- Three-week rule for any non-healing ulcer.

## 1 CHAPTER ONE Overview

Ulcers, patches and the tongue.

### 1.1 Approach

Take a history of duration, recurrence, pain, habits (tobacco, areca nut, alcohol), dentures, drugs, skin, eye and genital lesions, and general illness. Examine every subsite, palpate any lesion for induration, and feel the neck.

**CAUSES OF ORAL ULCERS**

| GROUP | EXAMPLES |
|---|---|
| Viral | Herpangina, primary and recurrent herpes simplex, hand, foot and mouth disease |
| Bacterial | Vincent infection (acute necrotising ulcerative gingivitis), tuberculosis, syphilis |
| Fungal | Candidiasis |
| Immune | Aphthous ulcers, Behçet syndrome |

| GROUP | EXAMPLES |
|---|---|
| Trauma | Cheek bite, jagged tooth, ill-fitting denture; chemical (aspirin, phenol, silver nitrate) and thermal burns |
| Neoplasm | Squamous cell carcinoma, minor salivary gland carcinoma, lymphoma |
| Skin disease | Erythema multiforme, lichen planus, pemphigus, mucous membrane pemphigoid, lupus |
| Blood disease | Leukaemia, agranulocytosis, cyclic neutropenia, pancytopenia |
| Other | Drug reactions, vitamin B12 and folate deficiency, radiation and chemotherapy mucositis, diabetes, uraemia |

**THE NON-HEALING ULCER**

An ulcer persisting beyond three weeks after any cause (a sharp tooth, a denture) has been removed, an indurated or rolled edge, a fixed lesion, numbness, or a hard neck node: biopsy for carcinoma.

CHAPTER TWO

# 2 Infections

Viruses, bacteria and fungi.

## 2.1 Viral

### 2.1.1 Herpangina

Coxsackie A infection of children: small vesicles on the pillars, tonsils, soft palate and uvula rupture into 2 to 4 mm ulcers with a yellow base and red areola, and heal within a week.

### 2.1.2 Herpetic gingivostomatitis

*Primary* herpes simplex in children: fever, malaise and clusters of vesicles anywhere in the mouth, with swollen, bleeding gums, healing in one to two weeks. *Recurrent* herpes in adults reactivates from the trigeminal ganglion after stress, fever, sunlight or immunosuppression, usually as *herpes labialis* at the vermilion

border, sometimes on the hard palate and attached gingiva. Treatment is symptomatic; aciclovir shortens recurrent attacks if started early.

#### 2.1.3 Hand, foot and mouth disease

A childhood viral infection with ulcers on the palate, tongue and cheeks and vesicles on the hands and feet.

## 2.2 Bacterial

#### 2.2.1 Vincent infection

Acute necrotising ulcerative gingivitis, caused by a fusiform bacillus and the spirochaete *Borrelia vincentii*: it begins at the interdental papillae, spreads along the gum margins under a grey necrotic slough, and smells foul; on the tonsil it is *Vincent angina*. A smear confirms it. Penicillin or erythromycin with metronidazole, mouthwashes and dental hygiene. In malnourished children it can spread into the face as *noma*.

#### 2.2.2 Specific infections

Tuberculosis (a painful, undermined ulcer, often on the tongue), syphilis and actinomycosis (sulphur granules) cause chronic ulcers; biopsy distinguishes them from cancer.

## 2.3 Fungal: candidiasis

*Candida albicans* takes hold in infants, diabetics, the immunosuppressed, denture wearers, after antibiotics, inhaled steroids or radiotherapy, and in a dry mouth.

- **Thrush (pseudomembranous):** white plaques that wipe off to leave a raw red base.
- **Erythematous (atrophic):** red patches, often under dentures.
- **Chronic hyperplastic (candidal leukoplakia):** a white patch that will not wipe off, usually at the commissures; it can harbour dysplasia, so biopsy it.
- **Angular cheilitis** at the corners of the mouth, and **median rhomboid glossitis** on the midline dorsum.

Topical nystatin or clotrimazole; fluconazole for severe disease; remove the cause.

ORAL HAIRY LEUKOPLAKIA

Corrugated white streaks on the lateral borders of the tongue that do not rub off, caused by Epstein–Barr virus in HIV or immunosuppression. It is harmless in itself but a marker of immune failure; it regresses with antiretroviral treatment.

# 3 CHAPTER THREE Immune and skin disorders

Ulcers that come back.

## 3.1 Aphthous ulcers

The commonest non-traumatic ulcers: recurrent, shallow, painful, on the *movable* mucosa (lips, cheeks, tongue, floor, soft palate) and sparing the hard palate and attached gingiva.



**FIGURE 1** Multiple aphthous ulcers on the uvula and faucial pillars (arrowheads).
Dhingra, fig. 43.1

- **Minor:** the common form, 2 to 10 mm, multiple, with a necrotic centre and red halo; heal in about two weeks without a scar.
- **Major:** over 1 cm, deep, often at the back of the mouth, lasting six weeks or more and healing with a scar.
- **Herpetiform:** crops of tiny ulcers.

The cause is unknown; associations include deficiency of vitamin B12, folate or iron, stress, hormones, food sensitivity, coeliac and Crohn's disease, and drugs such as

NSAIDs and nicorandil. Unlike viral ulcers, they recur, spare fixed mucosa and come without fever or nodes. Treat with topical steroids, chlorhexidine or tetracycline mouthwashes and topical anaesthetic gels, and correct deficiencies.

## 3.2 Behçet syndrome

Recurrent aphthous-like oral ulcers with genital ulcers and uveitis, sometimes with skin, joint and nervous system disease; the ulcers are punched out.

## 3.3 Erythema multiforme

A hypersensitivity reaction, often to herpes simplex or drugs (sulphonamides, anticonvulsants): bullae rupture into ulcers under a pseudomembrane, the lips form *haemorrhagic crusts*, and target lesions appear on the skin; the severe form merges into Stevens–Johnson syndrome. Supportive care; steroids for severe disease.

## 3.4 Lichen planus

A T-cell mediated disease. The *reticular* form shows lace-like white Wickham striae on both buccal mucosae and needs no treatment; the *erosive* form produces painful ulcers with a keratotic edge and responds to topical steroids. Erosive and atrophic forms carry a small risk of carcinoma, so they are followed up.

## 3.5 Pemphigus and pemphigoid

- **Pemphigus vulgaris:** autoantibodies to desmoglein 3 split the epithelium above the basal layer; fragile blisters break into painful erosions, often before skin lesions appear, in patients of fifty to seventy. Systemic steroids and immunosuppression.
- **Mucous membrane pemphigoid:** the split is beneath the epithelium; blisters on the gums, cheek and palate, and scarring of the conjunctiva. Steroids and dapsone.

## 3.6 Blood disorders and drugs

Leukaemia swells and ulcerates the gums; agranulocytosis and cyclic neutropenia cause ulceration as defences fail. A blood film, counts and marrow examination are needed.



**FIGURE 2** Palatal ulceration in a patient whose white cell count fell to 2,500 per cubic millimetre on cisplatin and fluorouracil chemotherapy.

Dhingra, fig. 43.4

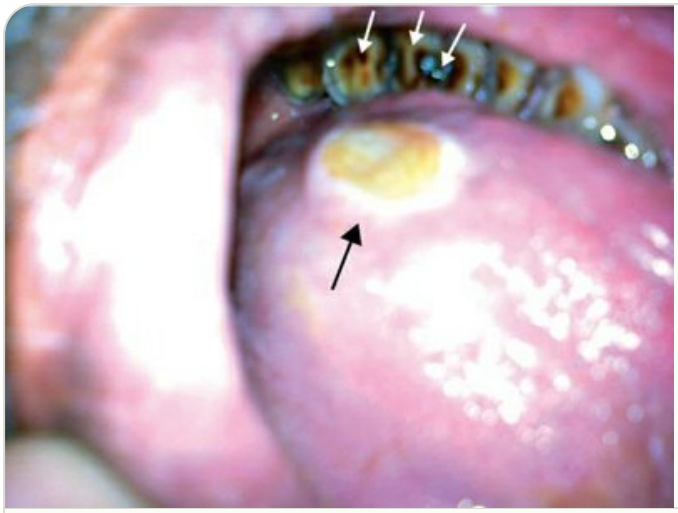
## 3.7 Mucositis

Radiotherapy and chemotherapy (methotrexate, fluorouracil, bleomycin) redden, then ulcerate the mucosa from about the second week; treatment is supportive.

# 4 CHAPTER FOUR Trauma

A sharp tooth can mimic cancer.

A jagged tooth or ill-fitting denture ulcerates the lateral tongue or cheek; a pencil or toothbrush injures the palate; acids, alkalis, hot fluids and an aspirin held against a painful tooth burn the mucosa.



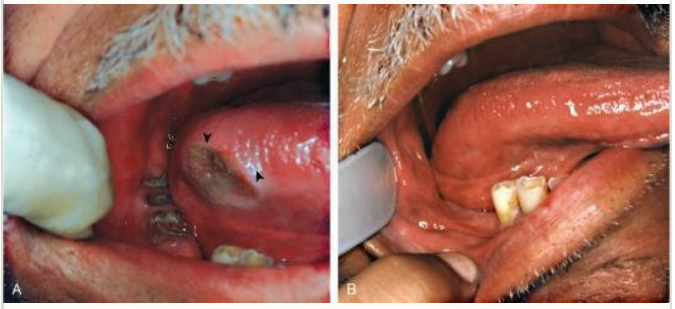



**FIGURE 3** First, an exophytic ulcerated mass on the left side of the tongue caused by sharp teeth (white arrows) in a man who chewed tobacco and pan; it healed after the teeth were extracted. Second, an ulcer on the lateral border of the tongue that simulated carcinoma (A) and healed completely after extraction of the jagged tooth (B).

Dhingra, figs 43.2 and 43.3

**REMOVE THE CAUSE, THEN REVIEW**

Smooth the tooth or adjust the denture and look again in two to three weeks. A traumatic ulcer heals; one that does not is a biopsy.

# 5 CHAPTER FIVE The tongue

Variants and deficiencies.

## 5.1 Variants

- **Geographic tongue (migratory glossitis):** red, smooth areas without papillae, outlined by a white keratotic margin, that change shape from day to day; harmless.
- **Fissured tongue:** congenital, or with deficiency states; with facial palsy and lip swelling it forms *Melkersson–Rosenthal syndrome*.
- **Hairy tongue:** elongated filiform papillae stained brown or black by bacteria, with smoking and antibiotics; scrape the tongue and treat the cause.
- **Median rhomboid glossitis:** a red, smooth, rhomboid area in front of the foramen caecum, now attributed to chronic candida.
- **Fordyce spots:** yellow ectopic sebaceous glands in the buccal and labial mucosa; normal.

- **Leukoedema:** a milky film on the buccal mucosa that disappears when the cheek is stretched, unlike leukoplakia.

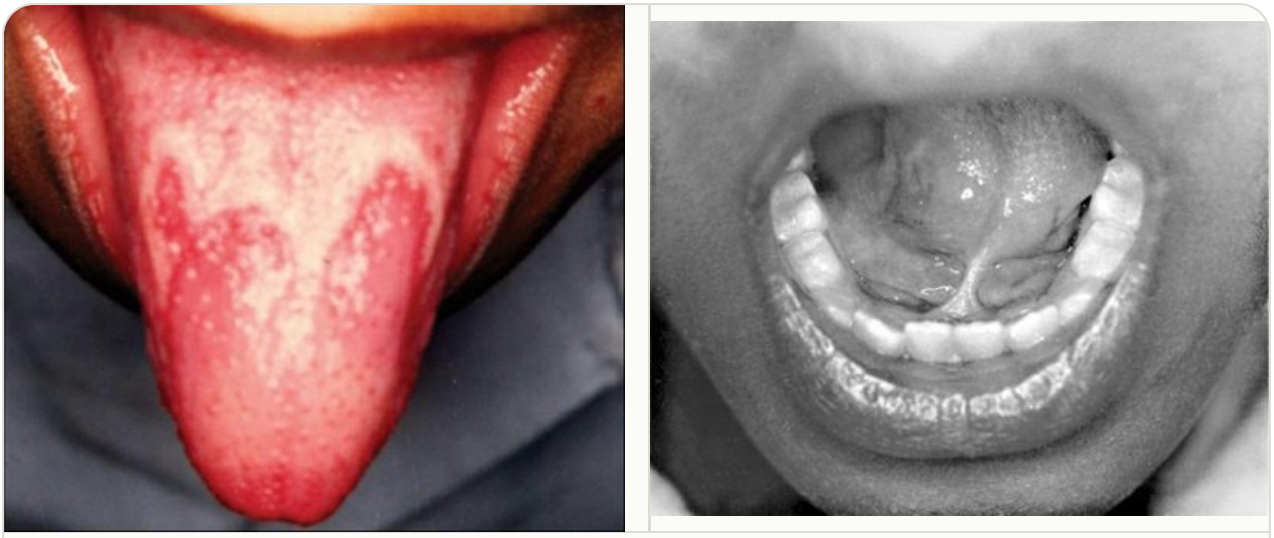


**FIGURE 4** First, geographic tongue. Second, tongue-tie.
Dhingra, figs 43.5 and 43.6

## 5.2 Tongue-tie (ankyloglossia)

A short frenulum. If the tongue can be protruded beyond the lower incisors it rarely affects speech; symptomatic ties are released (frenotomy, or transverse release with vertical closure).

## 5.3 Systemic clues

- **Smooth, red tongue:** iron, B12 or folate deficiency; ash-grey mucosa and atrophy in iron deficiency (Plummer–Vinson syndrome).
- **Angular cheilitis:** riboflavin or iron deficiency, candida.
- **Gingival hyperplasia and bleeding:** scurvy, leukaemia, phenytoin.
- **Strawberry tongue:** scarlet fever and Kawasaki disease; **Koplik spots:** measles.

## 5.4 Pigmentation

Racial melanosis, amalgam tattoo, Peutz–Jeghers syndrome, Addison disease, Kaposi sarcoma and heavy metals (the blue line of lead) all pigment the mouth; a

new or changing pigmented lesion, especially on the palate, is biopsied to exclude melanoma.

### 5.5 Macroglossia

Lymphangioma, haemangioma, hypothyroidism, acromegaly, amyloidosis, Down syndrome and storage diseases.

### 5.6 Nicotine stomatitis

Red pin-points in the centre of white papules on the palate of pipe and reverse smokers, from inflamed minor salivary gland ducts; it regresses when smoking stops. The heat, not the nicotine, is to blame.

## 6 CHAPTER SIX Submucous fibrosis

Areca nut and the mouth that will not open.

### 6.1 Definition and cause

Oral submucous fibrosis is a chronic, insidious fibrosis beneath the epithelium of the mouth and pharynx, first described in India by Joshi in 1953 and common across South and South-East Asia. *Areca (betel) nut* is the essential cause, chewed alone, with tobacco, or as pan; tobacco, alcohol, poor nutrition and an immune reaction to arecoline add to it.

### 6.2 Pathogenesis

Activated T-lymphocytes and macrophages release fibrogenic cytokines that drive fibroblasts to make more collagen while collagenase falls, so collagen is laid down faster than it is removed. The lamina propria becomes fibroelastic and avascular, the epithelium atrophies, and the fibrosis can reach the muscle.

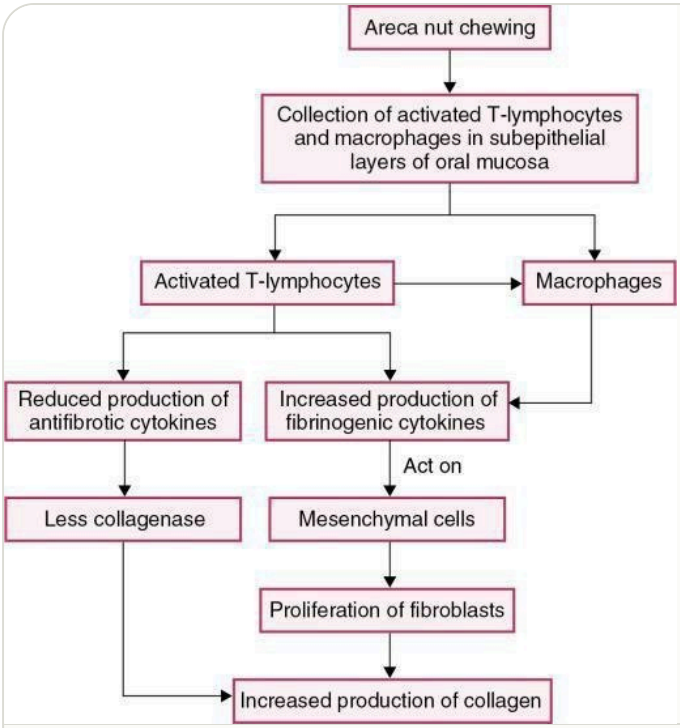



**FIGURE 5** The cellular immune response to areca nut that leads to increased collagen production in submucous fibrosis.

Dhingra, fig. 43.7

## 6.3 Clinical features

Usually between twenty and forty. Intolerance of chillies and spicy food, a burning mouth, recurrent vesicles on the palate and pillars, then difficulty in opening the mouth and protruding the tongue. The soft palate, pillars and buccal mucosa blanch and lose their suppleness, with palpable fibrous bands; trismus becomes progressive until the patient cannot brush the teeth.




**FIGURE 6** Submucous fibrosis: (A) blanching of the soft palate and faucial pillars; (B) marked trismus.

Dhingra, fig. 43.8

**A PREMALIGNANT CONDITION**

Carcinoma develops in about 3 to 7.6% (some series report more). Trismus makes examination difficult, so look carefully for leukoplakia and early cancer, and biopsy anything suspicious.

### 6.4 Treatment

- Stop areca nut, pan, tobacco and spicy food; correct anaemia and vitamin deficiency.
- Intralesional dexamethasone with hyaluronidase, given twice a week for eight to ten weeks, softens the bands and relieves trismus.
- Jaw-opening exercises.
- **Surgery** for advanced trismus: release of the bands with cover by skin graft, buccal fat pad, nasolabial or tongue flaps, a palatal island flap or radial forearm free flaps, with coronoidectomy and temporalis myotomy where needed; recurrence is common.

## 7 CHAPTER SEVEN Odontogenic cysts

Cysts in the jaws.

### 7.1 Dentigerous cyst

A developmental cyst around the crown of an unerupted or impacted tooth, usually the third molar or the upper canine, presenting in the second or third decade as a painless swelling of the jaw. Radiographs show a unilocular cyst containing the crown; in the maxilla the tooth may be pushed up to the orbital floor. Small cysts are removed with the tooth; large ones are first marsupialised to avoid a fracture. Ameloblastoma or carcinoma can arise in the wall, so the lining is always examined.

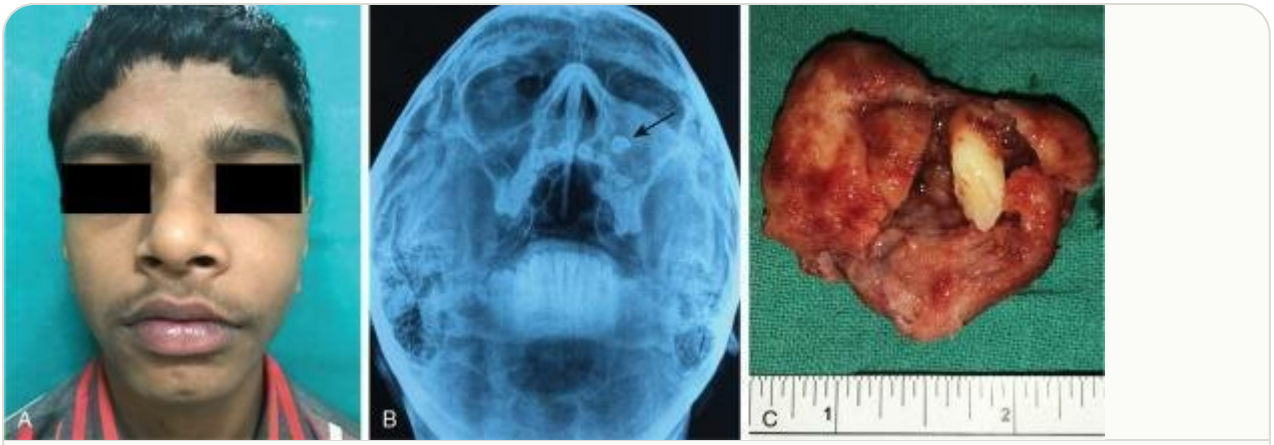



**FIGURE 7** Dentigerous cyst in a 14-year-old boy: (A) swelling of the left cheek; (B) radiograph; (C) the specimen with its tooth.

Dhingra, fig. 43.9

## 7.2 Dental (radicular) cyst

An inflammatory cyst at the apex (periapical) or side (lateral radicular) of a dead tooth. Enucleated when the tooth is extracted; lining left behind forms a residual cyst.

WHERE THE BOOKS DIFFER · WHERE THE BOOKS DIFFER

**Dhingra**

Groups the commoner ulcers, tongue lesions and submucous fibrosis in one chapter, with a strong Indian emphasis on areca nut and reverse smoking.

**Lee**

Adds the immunofluorescence patterns that separate pemphigus (intercellular "chicken-wire") from pemphigoid (linear basement membrane), and quotes a higher malignant risk for submucous fibrosis (7 to 30%).

The figures differ because the series differ; both treat submucous fibrosis as premalignant.

CHAPTER EIGHT

# 8 For the examination

What the examiner wants to hear.

## 8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Causes of oral ulceration** and the three-week rule.
- **Aphthous versus herpetic ulcers.**
- **Forms of candidiasis,** and why the hyperplastic form is biopsied.
- **Lichen planus,** pemphigus and pemphigoid.
- **Submucous fibrosis:** cause, features, premalignancy and treatment.
- **Dentigerous cyst.**

## 8.2 Common pitfalls

COMMON PITFALLS

- Prescribing repeated mouthwashes for a persistent single ulcer instead of a biopsy.
- Treating a white patch that will not rub off as thrush.
- Missing an early cancer in a mouth with submucous fibrosis and trismus.

Written for this app from Oxford Textbook of Otolaryngology ch. 54, 57, Oxford Specialist Handbook ch. 6, Oxford Handbook of ENT ch. 10, Dhingra ch. 43 and K. J. Lee ch. 36. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

SALIVARY GLANDS · INFECTION & INFLAMMATION

## Ranula of the Sublingual Gland

### IMAGING FINDINGS

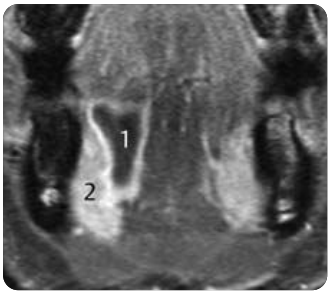
- Simple ranula: thin-walled cyst in the sublingual space
- Diving ranula: passes behind or through the mylohyoid into the submandibular space
- Tail sign pointing back to the sublingual space

### TOP DIFFERENTIALS

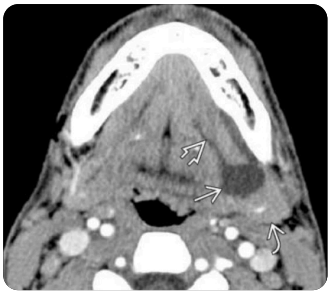
- Lymphatic malformation
- Dermoid or epidermoid
- Second branchial or thyroglossal cyst

### CHECKLIST

- Relation to the mylohyoid decides the approach
- Remove the sublingual gland to prevent recurrence




MRI, T1-weighted with gadolinium enhancement and fat suppression, coronal. There is a cystic lesion suggestive of a ranula (1), originating…



Axial CECT demonstrates a rounded cystic mass in the submandibular space abutting the posterior margin of the mylohyoid muscle and displacing…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 400.